



NATIONAL UNIVERSITY OF MODERN LANGUAGES

Web Engineering Project Report

NAME: M Abdul Rafay (FL23791)

Nazish Khan (FL23843)

Zohaib Ijaz (FL23811)

Ali Atif (FL23824)

PROGRAM: BSSE (5th Semester)

COURSE: Computer Networks

SUBMITTED TO: Mam Faryal

DATE: 9/27/2025

DAY: Saturday

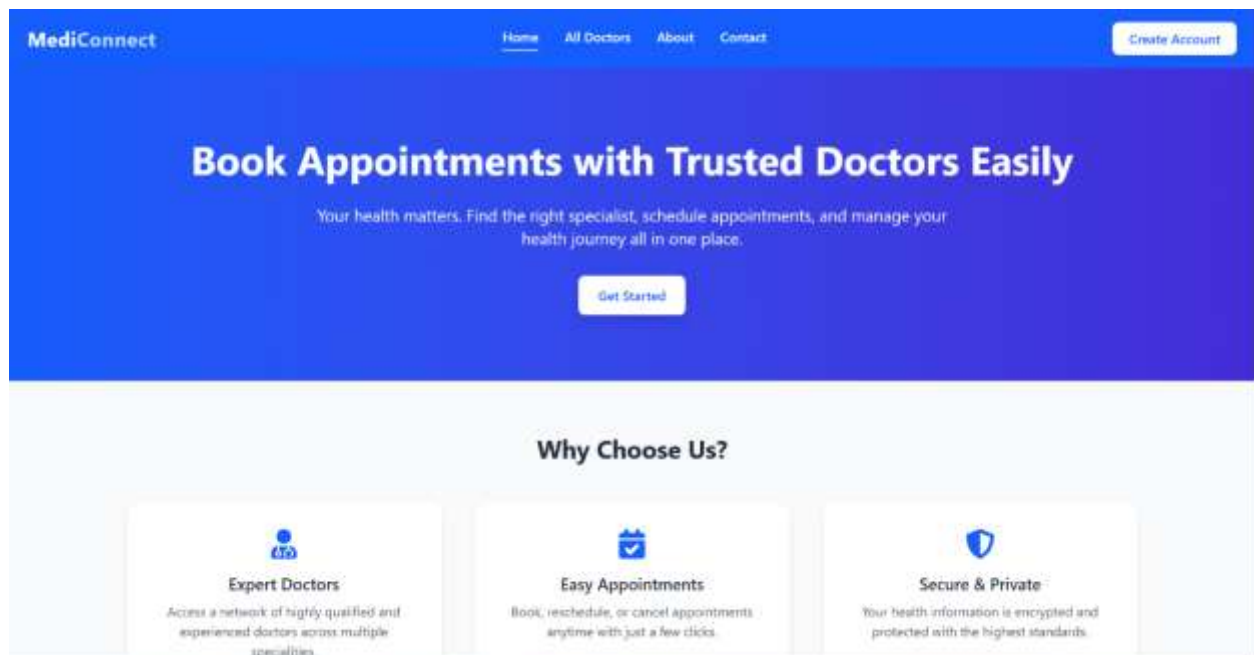
Contents

Doctor Appointment Booking System	3
Abstract.....	3
1. Introduction	3
2. Objectives	4
3. Problem Statement	4
4. Scope of the Project	4
5. Technology Stack	4
6. System Architecture	5
7. Use Case Diagram	5
8. Database Design Schema (ERD)	6
9. Functional Requirements	7
10. Non-Functional Requirement	7
11. API Design	7
12. Code Structure.....	8
14. Security Measures	8
15. User Interface Overview	8
17. Conclusion	12
19. Project Repository	12
<i>https://github.com/git-rafaykhan/doctor-oppointment-booking.git</i>	12

Doctor Appointment Booking System

Abstract

The Doctor Appointment Booking System is a full-stack web application designed to digitalize and automate the process of booking medical appointments. The system allows patients to view available doctors, select suitable time slots, book or cancel appointments, and manage their booking history. Administrators can manage doctors, schedules, and appointments through a centralized interface. This project demonstrates the application of modern web engineering principles, including client-server architecture, RESTful APIs, database design, authentication, and secure data handling.



- Figure 1: Home Page of the Doctor Appointment Booking System

1. Introduction

In traditional healthcare environments, appointment booking is often handled manually, leading to inefficiencies such as long waiting times, scheduling conflicts, and data management issues. This project aims to address these problems by providing a web-based solution that simplifies appointment scheduling for both patients and administrators.

2. Objectives

- To develop a secure and user-friendly doctor appointment booking platform
- To allow patients to book, view, and cancel appointments online
- To enable administrators to manage doctors and appointment schedules
- To apply core web engineering concepts in a real-world project

3. Problem Statement

Manual appointment booking systems are time-consuming, error-prone, and inefficient. Patients must physically visit or call clinics, while administrators struggle with record management. The Doctor Appointment Booking System provides an online solution to streamline scheduling, improve accessibility, and reduce operational overhead.

4. Scope of the Project

The scope of this project includes:

- User registration and authentication
- Doctor listing and specialization management
- Appointment booking and cancellation
- Admin panel for doctor and appointment management
- Secure data storage and retrieval

5. Technology Stack

Layer	Technology
Frontend	React.js
Backend	Node.js, Express.js
Database	Postgres
Authentication	JSON Web Tokens (JWT)
Version Control	Git and GitHub

6. System Architecture

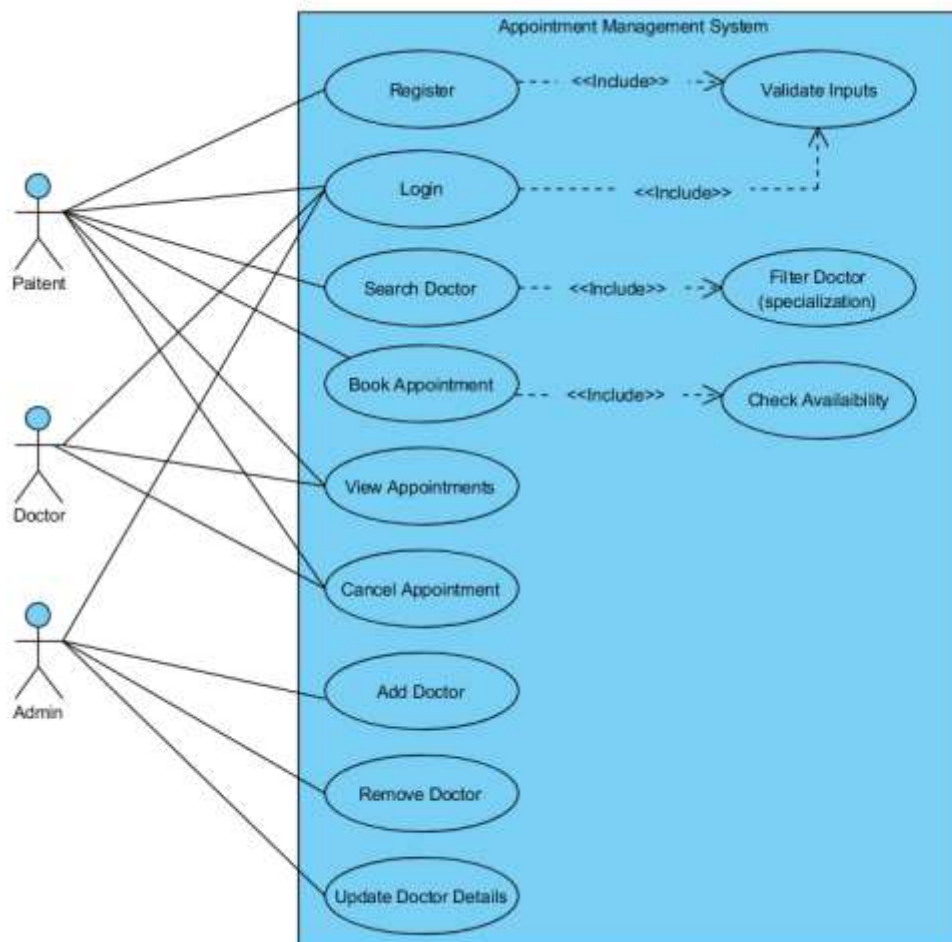
The application follows a client-server architecture. The frontend communicates with the backend using RESTful APIs. The backend handles business logic, authentication, and database operations, while MongoDB is used for persistent data storage.

7. Use Case Diagram

The system involves three main actors:

- Patient
- Doctor
- Administrator

Patients can register, log in, view doctors, book appointments, and cancel appointments. Administrators can add doctors, manage schedules, and view all appointments.



8. Database Design Schema (ERD)

The database consists of the following main entities:

- User
- Doctor
- Appointment

Each appointment links a patient with a doctor and a specific time slot.

```
model User {
  id      Int      @id @default(autoincrement())
  name    String
  email   String   @unique
  password String
  appointment Oppointment[]
}

model Doctor {
  id      Int      @id @default(autoincrement())
  name    String
  speciality String
  degree  String
  experience String
  about    String?
  fees     Float
  image    String
  createdAt DateTime @default(now())
  email    String   @unique
  password String
  address1 String   @default("not provided")
  address2 String   @default("not provided")
  appointment Oppointment[]
}

model Oppointment {
  id      String  @id @default(uuid())
  slotDate String
  userId  Int
  docId   Int
  amount  Float
  date    DateTime @default(now())
  cancelled Boolean @default(false)
  payment Boolean @default(false)
  iscompleted Boolean @default(false)
  slottime DateTime
  doc     Doctor   @relation(fields: [docId], references: [id])
  user    User     @relation(fields: [userId], references: [id])
}
```

9. Functional Requirements

- Users must be able to register and log in
- Users must be able to view available doctors
- Users must be able to book and cancel appointments
- Admin must be able to add and manage doctors
- Admin must be able to view all appointments

10. Non-Functional Requirement

- The system must be secure and protect user data
- The application should be responsive and user-friendly
- The system should provide fast response times
- The application should be scalable

11. API Design

Authentication APIs

- POST /api/v1/auth/register
- POST /api/v1/auth/login

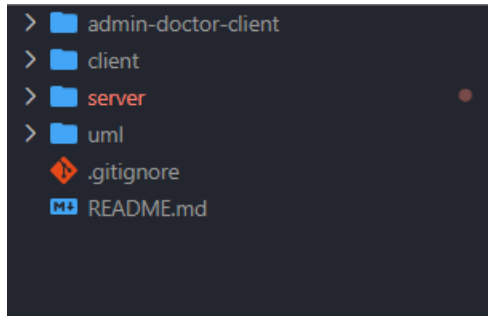
Doctor APIs

- GET /api/v1/doctors
- POST /api/v1/doctors (Admin only)

Appointment APIs

- POST /api/v1/appointments
- GET /api/v1/appointments
- DELETE /api/v1/appointments/:id

12. Code Structure



14. Security Measures

- JWT-based authentication
- Protected API routes
- Input validation
- Secure password storage

15. User Interface Overview

The user interface consists of the following screens:

- Login and Registration
- Doctor Listing
- Appointment Booking
- Appointment History
- Admin Dashboard

Welcome Back

Login to continue to MediConnect

Login

or



Continue with Google

Don't have an account? [Sign Up](#)

Create Account

Sign up to start your healthcare journey

Sign Up

or



Continue with Google

Already have an account? [Login](#)



Pookie
MBBS • 10+ Years
Pookie is a good doctor.
PKR 5000
Book Now
Pookie clinic
Pookie clinic opposite



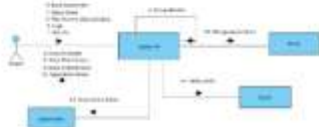
DR. Saddle
MBBS • 10+ Years
Saddle is specialized in killing people.
PKR 4500
Book Now
Saddle Hospital
Saddle Hospital 2



DR Umer Raja
MBBS • 10+ Years
Dr umer raja is a great doctor, specialized in heart surgery.
PKR 2000
Book Now
Khaman main city
Raja clinic



DR Zohaib Ijaz
MBBS • 10+ Years
Dr zohaib is a good doctor.
PKR 5000
Book Now
Ratge Road
Santal clinic



xyz
mbbs • 5 Years
kjhghkjh
PKR 1222
Book Now
xzt
kjhghkjh



DR Zohaib Ijaz

Neurologist

PKR 5000

Dr zohaib is a good doctor.

Book an Appointment

Select Date:

12/27/2025



Available Time Slots:

09:00 AM

10:00 AM

11:00 AM

01:00 PM

02:00 PM

03:00 PM

04:00 PM

Confirm Appointment



Upload profile picture

Full Name

Email

Password

Speciality

Select Speciality

Degree

Experience

Select Experience

Fees (PKR)

Clinic / Hospital Address (Line 1)

Address Line 2 (Optional)

About Doctor

Add Doctor

Tables

production

neondb

Database studio

public

Search

_prisma_migrations

Doctor

Appointment

User

20 rows • 443 rows

50

0

Filters

Columns

+ Add record

#	name	specialty	degree	experience
1	Dr. Richard James	Pediatrician	MD&C	5 Years
2	Dr. Emily Larson	Dermatologist	MD&C	2 Years
3	Dr. Sarah Patel	Neurologist	MD&C	6 Years
4	Dr. Christopher Lee	Neurologist	MD&C	2 Years
5	Dr. Jennifer Garcia	Dermatologist	MD&C	1 Year
6	Dr. Andrew Williams	Orthopedic	MD&C	10+ Years
7	Dr. Chris Davis	Orthopedic	MD&C	6 Years
8	Dr. Timothy White	Pediatrician	MD&C	6 Years
9	Dr. Ava Mitchell	Dermatologist	MD&C	5 Years
10	Dr. Jeffrey King	Orthopedic	MD&C	5 Years
11	Dr. Zoe Kelly	Dermatologist	MD&C	10+ Years
12	Dr. Patrick Harris	Neurologist	MD&C	6 Years
13	Dr. Chloe Evans	Cardiologist	MD&C	10+ Years
14	Dr. Ryan Martinez	Cardiologist	MD&C	5 Years
15	Dr. Amelia Hill	Pediatrician	MD&C	2 Years
16	Pookie	Cardiologist	MD&C	10+ Years
17	Dr. Eddie	Pediatrician	MD&C	10+ Years
18	Dr. Umar Raja	Cardiologist	MD&C	10+ Years
19	Dr. Zohair Ijaz	Neurologist	MD&C	10+ Years
20	aya	Dermatologist	MD&C	5 Years

17. Conclusion

The Doctor Appointment Booking System successfully demonstrates the implementation of a real-world web application using modern web technologies. The project enhances understanding of full-stack development, database design, API development, and secure authentication. Future improvements may include payment integration, email notifications, and doctor availability management.

19. Project Repository

The complete source code of the Doctor Appointment Booking System is available on GitHub at the following link:

<https://github.com/git-rafaykhan/doctor-appointment-booking.git>

The repository contains the full frontend and backend implementation, database models, API routes, and project documentation.